

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Advanced Machine Learning, Performance Engineering, Advanced NLP, Optimization in CS

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Java, Go, JavaScript, TypeScript, C++, SQL, Bash

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, client/server, async services, multi-tenant architecture, integration testing

Data & ML

Spark / PySpark, data pipelines / ETL, PyTorch, LLM integration, RAG, pandas, NumPy

Databases

MongoDB, PostgreSQL, Redis, SQLite, DynamoDB

Cloud & DevOps

Docker, Git, GitHub Actions, CI/CD, Linux, AWS, GCP, Azure

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions** in a **regulated clinical healthcare** setting, integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Architected a full-stack **internal tool** (React, Node.js, MongoDB) used daily by the AI team to organize and label data for **production ML models**, replacing spreadsheet workflows with secure role-based REST APIs; cut manual work by **100+ hours/month** and improved throughput by **60%**.
- Acted as **primary technical point of contact** for the eTriage backend, owning it end-to-end across feature releases and client integrations; collaborated cross-functionally, monitored production logs to triage incidents, hardened authentication and rate limiting, and ran knowledge transfer to onboard a second developer.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team, in a fast-moving startup environment.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories and integrated **LLMs** into a chat product for context-aware replies, iterating on prompt design and response quality.

Projects

AgenticSearch - AI/ML Integrated into a Backend Data Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a **Python/FastAPI service** that integrates **LLMs** into a multi-stage data pipeline: plans typed retrieval, reranks pages, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every value.
- Hardened the service with **150+ automated tests**, provider routing/fallback across LLM backends, and pipeline instrumentation to keep model integration predictable across failures.

Spark ETL Optimization - Large-Scale Data Pipeline

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI, SQL

- Built and optimized a Spark/PySpark **data pipeline** on a **~1.4B-row** dataset, cutting end-to-end ETL runtime **~25%** (17 to 19 min down to 14.3 min) via broadcast joins, Adaptive Query Execution, skew-join handling, and partition/column pruning.
- Profiled stage-level timings in the Spark UI to localize shuffle and skew bottlenecks and document a repeatable optimization playbook, reducing the tail-latency ratio (P99/P50) from **2.14x to 1.74x**.

Underdogs Fitness - 0→1 Production Full-Stack App (Live 3+ Years)

underdogsfitness.in

MongoDB Atlas, Express, React + Redux, Node.js, Stripe

- Shipped a **0→1 production full-stack app** solo end-to-end (React + Redux, Node.js/Express): auth, membership and attendance, **Stripe payments with a partial-payment ledger**, and cron automation, **live and actively used 3+ years**.
- Designed it multi-branch from day 1; when the client expanded a year later it onboarded as a **config change with zero schema migration**, deployed via git-webhook automation.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.